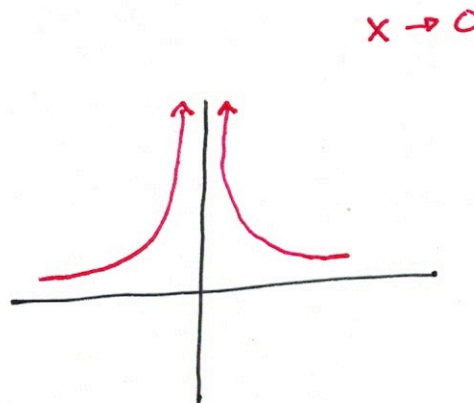


Section 2.2 Limits of a Function

Ex ① $f(x) = \frac{1}{x^2}$ what happens when x is close to zero?

x	$f(x)$
-0.2	25
-0.1	100
-0.01	10,000
0	undefined
0.01	10,000
0.1	100
0.2	25

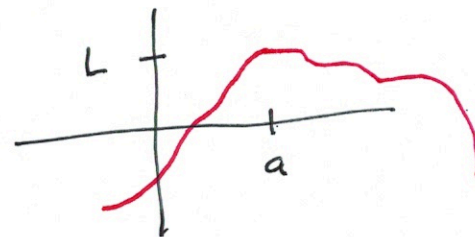


$$\lim_{x \rightarrow 0} \frac{1}{x^2} = \boxed{\infty}$$

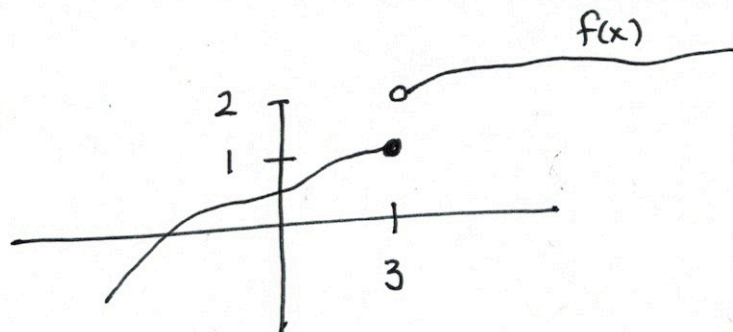
NOTE If $\lim_{x \rightarrow a} f(x) = L$

then $\lim_{x \rightarrow a^-} f(x) = L$

and $\lim_{x \rightarrow a^+} f(x) = L$



Ex ②



$$\lim_{x \rightarrow 3^-} f(x) = 1 \quad \text{and} \quad \lim_{x \rightarrow 3^+} f(x) = 2$$

$\lim_{x \rightarrow 3} f(x)$ Does Not Exist
(DNE)

$$\text{Ex } ③ \quad \lim_{x \rightarrow 0} \frac{5}{1+e^x} = \frac{5}{1+e^0} = \frac{5}{1+1} = \frac{5}{2} = \boxed{2.5}$$

$$\text{Ex } ④ \quad \lim_{x \rightarrow \infty} \frac{5}{1+e^x} \Rightarrow \frac{5}{1+e^\infty} = \frac{5}{\infty} = \boxed{0}$$

Ex (5) $\lim_{x \rightarrow 0} \frac{\sin x}{x}$ use a table of values to estimate the limit

$$\boxed{\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1}$$

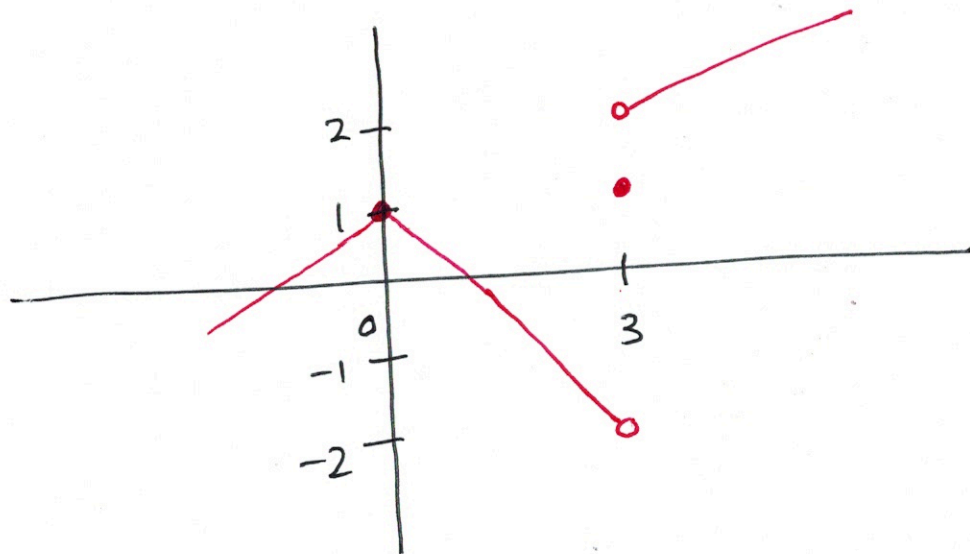
x	y
-.03	1.0
-.02	1.0
-.01	1.0
0	undef.
.01	1.0
.02	1.0
.03	1.0

$$\begin{aligned} \text{Ex (6)} \quad \lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3} &= \lim_{x \rightarrow 3} \frac{(x+3)(\cancel{x-3})}{\cancel{x-3}} \\ &= 3+3 = \boxed{6} \end{aligned}$$

Ex ⑦ Graph $f(x)$ with

$$f(0) = -1, \quad f(3) = 1$$

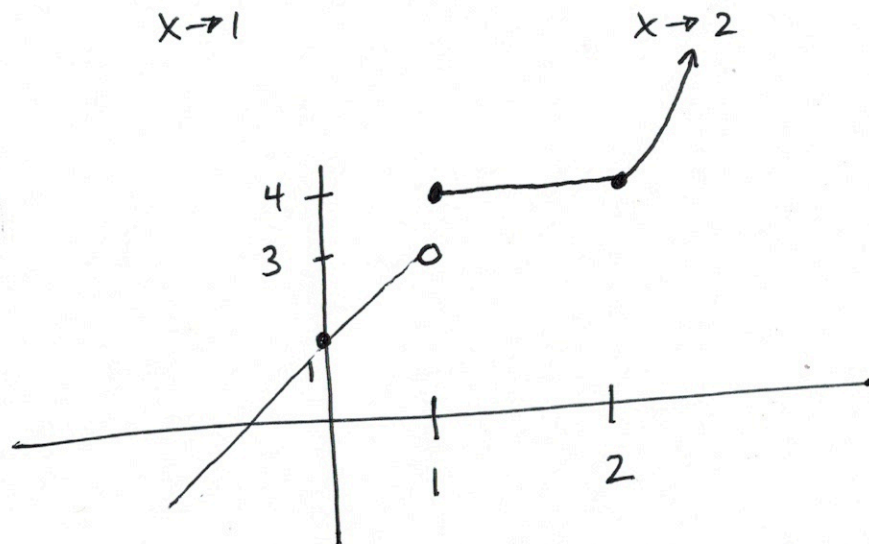
$$\lim_{x \rightarrow 0} f(x) = 1, \quad \lim_{x \rightarrow 3^-} f(x) = -2, \quad \lim_{x \rightarrow 3^+} f(x) = 2$$



Ex (8)

$$f(x) = \begin{cases} 2x+1 & \text{if } x < 1 \\ 4 & \text{if } 1 \leq x \leq 2 \\ x^2 & \text{if } x > 2 \end{cases}$$

Find $\lim_{x \rightarrow 1} f(x)$ and $\lim_{x \rightarrow 2} f(x)$



$\lim_{x \rightarrow 1} f(x)$ DNE.

$$\lim_{x \rightarrow 2} f(x) = 4$$